Proof. We will show the result for small t; by analyticity of the family, it then follows for all t. For convenience, denote by $\mathcal{M}_c$ the factor $\mathcal{M}_{g_i,n_i}$. Since ∂V_0 is positive dimensional, it must vary on some other factor $\mathcal{M}_v$. Going forward, we will label boundary strata by the multi-curve (considered up to action of the mapping class group) pinched, rather than the dual graph. Let Λ be the multi-curve corresponding to the dual graph Γ, and then denote by Δ_Λ the boundary stratum Δ_Γ.

Fix a point $X \in \partial V_0$. The image of the projection of $\pi_v : \partial V_0 \to \mathcal{M}_v$ is a totally geodesic subvariety ([BDR24, Theorem 1.4] or [AW24, Theorem 1.3]) of positive dimension, so it is non-compact (e.g. by using the $GL_2(\mathbb{R})$ action). Consider a degeneration in this factor (which may lead to degeneration in other factors, but not in $\mathcal{M}_c$, since it is 0-dimensional there). This lands at X' in some more degenerate boundary stratum $\Delta_{\Lambda\cup\Lambda'}$, obtained from Δ_Λ by further pinching some multi-curve Λ'. By [AW24, Theorem 1.6 (4)], Λ is a union of disjoint equivalence classes (in their sense), since X is in closure of V_0, and Λ is the multi-curve pinched at X. And so is $\Lambda \cup \Lambda'$, since X' is in closure of V_0. It follows that Λ' is also a union of disjoint equivalence classes. Then by [AW24, Corollary 3.17], nearby X' we can unpinch the nodes corresponding to Λ. This means that locally near X', the boundary of $\bar{V}_0 \cap \Delta_{\Lambda'}$ in $\Delta_{\Lambda\cup\Lambda'}$ equals $\bar{V}_0 \cap \Delta_{\Lambda\cup\Lambda'}$.

Now we consider other elements V_t of the family. By (i), we get a family in Δ_Λ deforming the boundary of V_0 there. Similarly, we get a family in $\Delta_{\Lambda\cup\Lambda'}$ and one in $\Delta_{\Lambda'}$, each deforming the boundaries of V_0 in those respective strata. By (ii), the relevant nodes can be smoothed at other elements of the family, locally near X', i.e. for a small open neighborhood U of X',

$$\overline{\bar{V}_t \cap \Delta_\Lambda} \cap \Delta_{\Lambda\cup\Lambda'} \cap U = \bar{V}_t \cap \Delta_{\Lambda\cup\Lambda'} \cap U. \tag{1}$$

and, similarly (now also using the result of the previous paragraph), we get

$$\overline{\bar{V}_t \cap \Delta_{\Lambda'}} \cap \Delta_{\Lambda\cup\Lambda'} \cap U = \bar{V}_t \cap \Delta_{\Lambda\cup\Lambda'} \cap U. \tag{2}$$

Note that $\mathcal{M}_c$ is also a factor in $\Delta_{\Lambda\cup\Lambda'}$, since the degeneration corresponding to Λ' does not affect $\mathcal{M}_c$. And the image of the projection $\pi_c : \bar{V}_0 \cap \Delta_{\Lambda\cup\Lambda'} \to \mathcal{M}_c$ is identical to image of the projection $\pi_c : \bar{V}_0 \cap \Delta_\Lambda \to \mathcal{M}_c$. In $\Delta_{\Lambda'}$, $\mathcal{M}_c$ undegenerates into some factor $\mathcal{M}_{c'}$, i.e. $\mathcal{M}_c$ a factor in $\partial\mathcal{M}_{c'}$. Now the image of the projection $\pi_{c'} : \bar{V}_0 \cap \Delta_{\Lambda'} \to \mathcal{M}_{c'}$ has dimension one (the dimension is ≥ 1 since it admits a degeneration, to $\mathcal{M}_c$, by pinching Λ; the dimension is ≤ 1 is since it lies on the boundary of the surface V_0). Again by [BDR24, Theorem 1.7] or [AW24, Theorem 1.3], the projection of $\bar{V}_0 \cap \Delta_{\Lambda'}$ to $\mathcal{M}_{c'}$ is totally geodesic. By [McM09], this is holomorphically rigid. In particular, $\pi_{c'}(\bar{V}_t \cap \Delta_{\Lambda'})$ does not depend on t, for t in some small disc. But $\pi_c(\bar{V}_t \cap \Delta_{\Lambda\cup\Lambda'})$ is (locally, near X') the boundary of this, since by (2), this is true of the whole product. We conclude that $\pi_c(\bar{V}_t \cap \Delta_{\Lambda\cup\Lambda'})$ also does not depend on t.

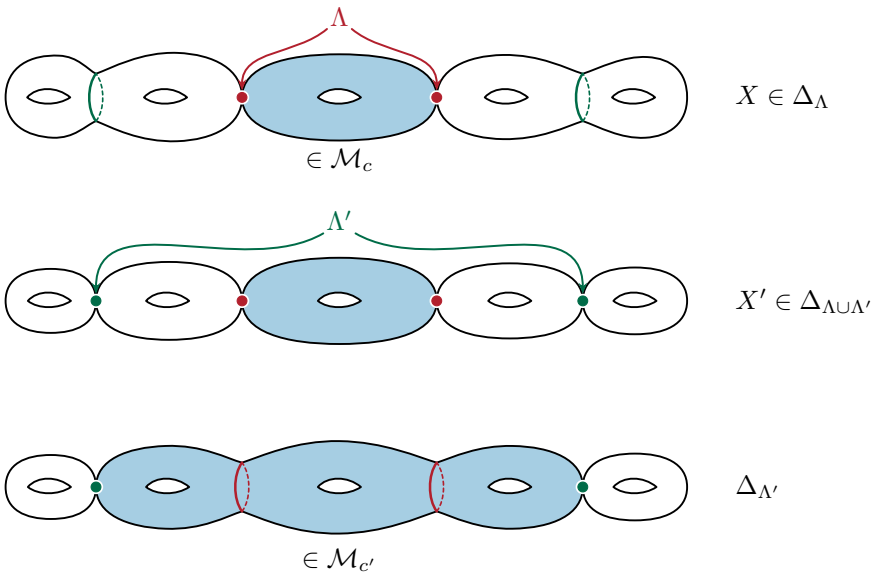


Figure 1: Degenerations used in the proof.

Now we are assuming for each t that $\pi_c(\bar{V}_t \cap \Delta_\Lambda)$ consists of a single point, call it Y_t. Since Λ' does not contain any curves on the $\mathcal{M}_c$ factor, and using (1), we get that $\pi_c(\bar{V}_t \cap \Delta_{\Lambda\cup\Lambda'})$ also must equal $\{Y_t\}$. And since we saw in the previous paragraph that $\pi_c(\bar{V}_t \cap \Delta_{\Lambda\cup\Lambda'})$ does not depend on t, neither does Y_t.

■

References

[AW24] Francisco Arana-Herrera and Alex Wright, *The geometry of totally geodesic subvarieties of moduli spaces of Riemann surfaces*, arXiv e-prints (2024), arXiv:2412.16330. 1

[BDR24] Frederik Benirschke, Benjamin Dozier, and John Rached, *The boundary of a totally geodesic subvariety of moduli space*, arXiv e-prints. To appear in Journal of the European Math. Society. (2024), arXiv:2412.06765. 1

[McM09] Curtis T. McMullen, *Rigidity of Teichmüller curves*, Math. Res. Lett. **16** (2009), no. 4, 647–649. MR 2525030 1